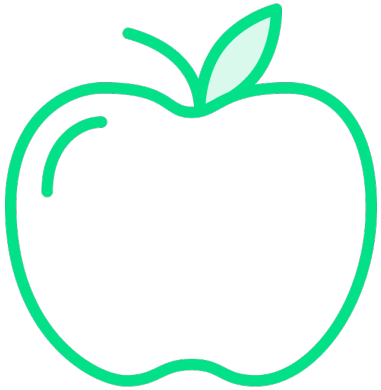
Types of Custom Vision Model

John Savill
Chief Architect

@ntfaqguy | onboardtoazure.com

What is Custom Vision?



The Azure AI services have a number of vision pre-trained models that are useful in many situations.

You may need image recognition for your own classification or object detection needs.

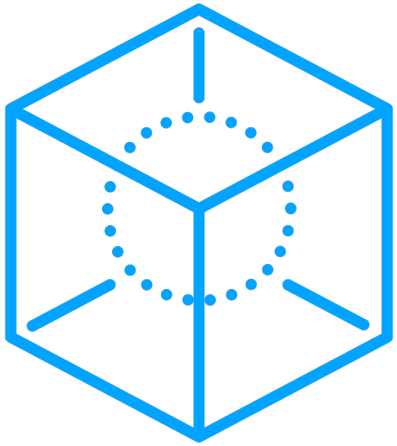
Custom vision enables us to train the model using our data.

Once trained it can be used for prediction.

Custom Vision vs Image Analysis 4.0

Custom Vision Service	Image Analysis 4.0 Service
Uses convolution neural network (CNN) base model	Uses a multi-modal transformer base model
Minimum of 15 images per category for training data	Minimum of 2-5 images per category for training data
Data must be uploaded to the service	Data read from users blob storage account

Creating the Resource for Custom Vision



Because we will use the Image Analysis 4.0 based customized vision we do not use “Custom Vision” resource.

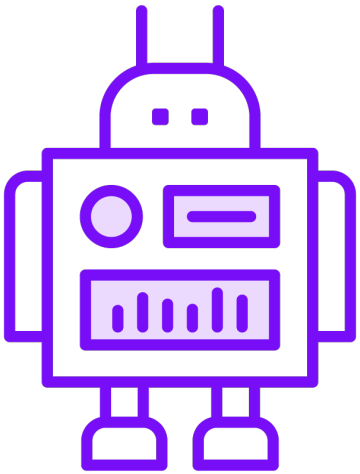
A regular computer vision or Azure AI multi-service resource is utilized.

Preparing Your Training Dataset

John Savill
Chief Architect

@ntfaqguy | onboardtoazure.com

Training Data



To train a custom model we need training data.

The training data should cover the complete types of object that need to be identified.

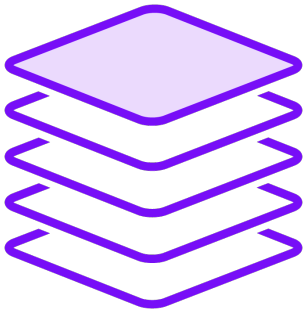
We will provide training data that has been pre-labeled for image classification OR object detection.

Azure Machine Learning workspaces provides an intuitive environment for data labelling.

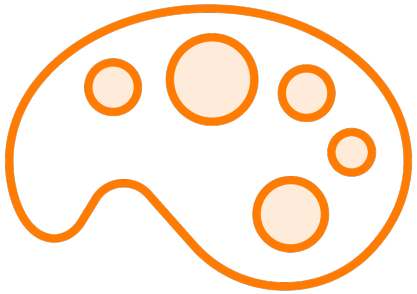
This will generate a COCO file that will be used for the actual model training.

Attributes of Good Training Data

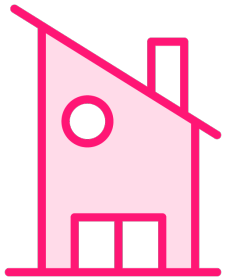
Balance



Quantity



Variety



Partial images

Demo: Creating a Training Dataset

John Savill
Chief Architect

@ntfaqguy | onboardtoazure.com

Key Steps

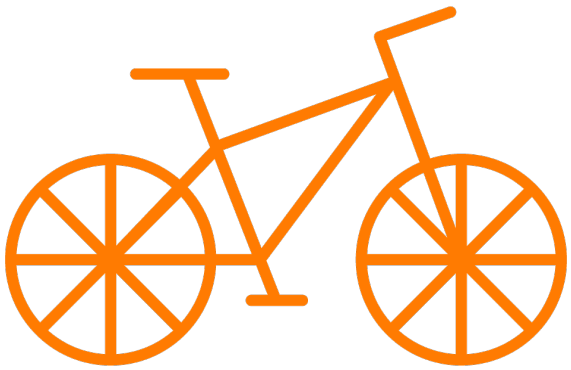
- Upload training data to blob storage
- Create a new computer vision custom model
- Create a new dataset using the uploaded blob content
- Label the data in Azure AI Machine Learning studio
- Generate and store the COCO file

Training and Publishing a Custom Vision Model

John Savill
Chief Architect

@ntfaqguy | onboardtoazure.com

Training a Custom Model



With the training data labelled we can train the custom model.

A separate set of trained data can be used for evaluation.

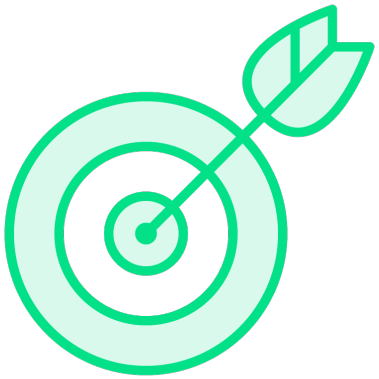
A budget of maximum hours is allocated for the training effort.

Training is financially expensive so be careful and set appropriate budgets.

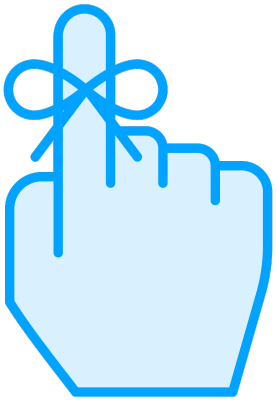
Image Analysis 4.0 models can take relatively longer to train than previous models.

Training can be an iterative process.

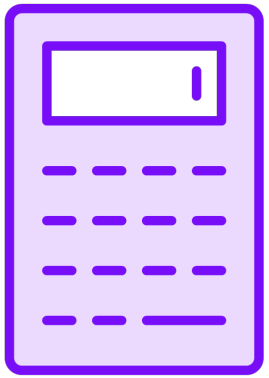
Model Performance Metrics



Precision

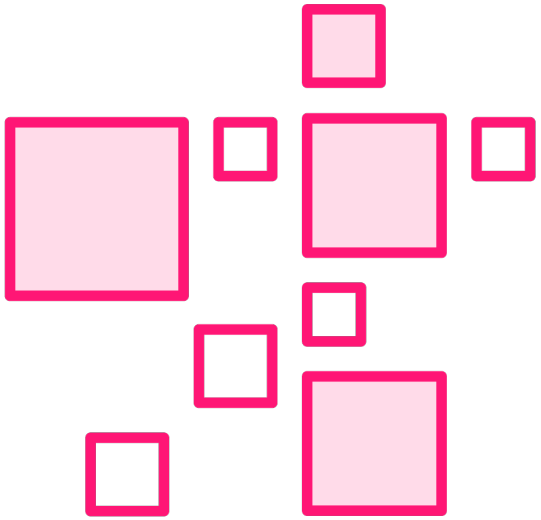


Recall

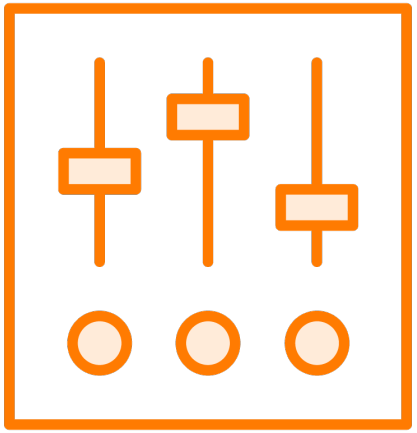


Mean Average Precision (mAP)

Model Performance Metrics

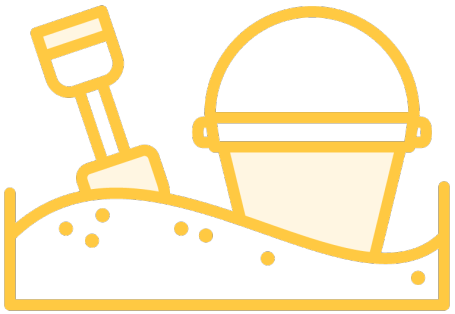


Intersection over Union (IoU)



Probability Threshold

Using Your Custom Model



Custom models can be tested within Vision Studio.

Once satisfied with the model performance it can be utilized by the regular imageanalysis:analyze API call.

Specify the custom model name instead of the features query parameter.

Demo: Training a Custom Vision Model

John Savill
Chief Architect

@ntfaqguy | onboardtoazure.com

Key Steps

Create a new custom model and pass in training data

Wait for the training to complete

Perform testing of performance and improve as needed

Utilize the custom model using standard APIs specifying the custom model name

Implementing Custom Vision Models Review

John Savill
Chief Architect

@ntfaqguy | onboardtoazure.com

Summary

While there are two types of custom image model available, we will normally use the Image Analysis 4.0 version.

Training data should be of sufficient volume, balance and variety to enable the trained model to work in real world situations.

Model training is commonly an iterative process.

The regular API is used but the custom vision model name specified instead of a list of required features.